WISP: Wildfire Information and Sensor Planning

Mathematical Formulation

Abstract

WISP (Wildfire Information and Sensor Planning) is a real-time system for autonomous drone-based wildfire monitoring. It couples a GPU-accelerated ensemble fire-spread model (Rothermel 1972 surface ROS + Van Wagner 1977 crown fire, propagated via a level-set PDE) with a Gaussian Process environmental prior, an Ensemble Kalman Filter for data assimilation, and an information-theoretic greedy orienteering planner that routes drones to maximally reduce uncertainty about fire arrival times. This document derives every mathematical component of the system.

1 System Overview

Let $\mathcal{G} = \{(r, c) : 0 \le r < R,\ 0 \le c < C\}$ be a $R \times C$ grid at resolution Δx m. The hidden state consists of:

- $\phi^*(r, c, t)$ – the signed-distance field (SDF) of the true fire front,
- $\theta^*(r, c)$ – static terrain (elevation, slope, aspect, fuel model),
- $m^*(r, c, t)$ – dead fuel moisture content (FMC),
- $\mathbf{u}^*(r, c, t)$ – surface wind (speed s, direction d).

At each planning cycle (interval ΔT_{cyc}) the system:

1. Fits a GP to accumulated sensor observations to produce posterior fields $(\hat{m}, \hat{s}, \hat{d})$,
2. Runs an N-member ensemble of the level-set fire model to obtain arrival-time distributions,
3. Computes an information field $w(r, c)$ weighting cells by expected variance reduction,
4. Plans drone paths that maximise $\sum w$ subject to range constraints.

2 Rothermel Surface Rate of Spread

The Rothermel (1972) model predicts the no-wind, no-slope rate of spread R_0 and its amplification by wind and slope. All fuel parameters are taken from the Scott–Burgan 40 (SB40) fuel catalog.

2.1 Fuel parameters

For each SB40 fuel type the catalog provides:

w_0	oven-dry fuel loading (kg/m^2),
σ	surface-area-to-volume ratio (SAV, m^{-1}),
δ	fuel bed depth (m),
M_x	moisture of extinction (fraction),
h	heat of combustion (kJ/kg).

Internal computations follow Rothermel's original SI convention; the SAV is converted to ft^{-1} where required: $\sigma_{\text{ft}} = \sigma/3.2808$.

2.2 Packing ratio

$$\rho_p = 513 \text{ kg/m}^3 \quad \text{(particle density)},$$

$$\beta = \frac{w_0}{\delta\, \rho_p}, \tag{1}$$

$$\beta^* = 3.348\, \sigma_{\text{ft}}^{-0.8189}, \tag{2}$$

$$\hat{\beta} = \beta/\beta^*. \tag{3}$$

2.3 Reaction intensity

$$A = 133\,\sigma_{\mathrm{ft}}^{-0.7913}, \tag{4}$$

$$\Gamma_{\max} = \frac{\sigma_{\mathrm{ft}}^{1.5}}{495 + 0.0594\,\sigma_{\mathrm{ft}}^{1.5}}, \tag{5}$$

$$\Gamma = \Gamma_{\max}\,\hat{\beta}^{A}\,\exp\bigl(A(1-\hat{\beta})\bigr), \tag{6}$$

$$w_n = w_0\,(1 - s_t), \quad s_t = 0.0555, \tag{7}$$

$$\eta_s = \min\bigl(0.174\,s_e^{-0.19},\,1\bigr), \quad s_e = 0.01, \tag{8}$$

$$r_m = \min\bigl(m/M_x,\,1\bigr), \tag{9}$$

$$\eta_m = \max\bigl(0,\;1 - 2.59\,r_m + 5.11\,r_m^2 - 3.52\,r_m^3\bigr), \tag{10}$$

$$I_R = \Gamma\,w_n\,h\,\eta_m\,\eta_s \quad [\mathrm{kW/m^2}], \tag{11}$$

where m is the local FMC (fraction).

2.4 Propagating flux ratio

$$\xi = \frac{\exp\bigl[(0.792 + 0.681\sqrt{\sigma_{\mathrm{ft}}})(\beta + 0.1)\bigr]}{192 + 0.2595\,\sigma_{\mathrm{ft}}}. \tag{12}$$

2.5 Wind and slope amplification

The midflame wind speed U_f (ft/min) is obtained from the 10-m wind speed U_{10} (m/s) via a wind adjustment factor f_w:

$$U_f = U_{10}\,f_w \times 196.85, \tag{13}$$

where $f_w \in [0.4,\,0.6]$ depends on canopy cover fraction.

Wind factor:

$$C = 7.47\exp(-0.133\,\sigma_{\mathrm{ft}}^{0.55}), \tag{14}$$

$$B = 0.02526\,\sigma_{\mathrm{ft}}^{0.54}, \tag{15}$$

$$E = 0.715\exp(-3.59\times 10^{-4}\,\sigma_{\mathrm{ft}}), \tag{16}$$

$$\phi_w = C\,U_f^{B}\,\hat{\beta}^{-E}. \tag{17}$$

Slope factor:

$$\phi_s = 5.275\,\beta^{-0.3}\tan^2\alpha, \tag{18}$$

where α is the terrain slope angle.

2.6 Vector combination and head-fire direction

Wind and slope forces are resolved as 2-D vectors:

$$r_x = \phi_w\sin\psi_w + \phi_s\sin\psi_s, \tag{19}$$

$$r_y = \phi_w\cos\psi_w + \phi_s\cos\psi_s, \tag{20}$$

$$\phi_{\mathrm{comb}} = \sqrt{r_x^2 + r_y^2}, \tag{21}$$

$$\theta_{\max} = \mathrm{atan2}(r_x, r_y) \mod 360^\circ, \tag{22}$$

where $\psi_w = (\psi_{\mathrm{wind}} + 180^\circ) \mod 360^\circ$ and ψ_s is the upslope aspect.

2.7 Head-fire ROS

$$Q_{\mathrm{ig}} = 250 + 1116\,m \quad [\mathrm{cal/g}], \tag{23}$$

$$\varepsilon = \exp(-138/\sigma_{\mathrm{ft}}), \tag{24}$$

$$\rho_b = w_0/\delta, \tag{25}$$

$$R_{\mathrm{head}} = \frac{I_R\,\xi\,(1 + \phi_{\mathrm{comb}})}{\rho_b\,\varepsilon\,Q_{\mathrm{ig}} \times 60} \quad [\mathrm{m/s}]. \tag{26}$$

3 Directional Rate of Spread

Spread is modelled as an Anderson (1983) ellipse. The length-to-breadth ratio and eccentricity are:

$$U_{\mathrm{mid}} = U_{10}\,f_w \times 3.6 \quad [\mathrm{km/h}], \tag{27}$$

$$L/B = \min\bigl(\max(1 + 0.25\,U_{\mathrm{mid}},\,1),\,8\bigr), \tag{28}$$

$$e = \frac{L/B - 1}{L/B + 1}. \tag{29}$$

For a local outward normal $\hat{\mathbf{n}}$ making angle θ with the head-fire direction:

$$R(\theta) = R_{\mathrm{head}}\,\frac{1 - e}{1 - e\cos\theta}. \tag{30}$$

The angle θ is computed from the level-set gradient $\nabla\phi = (g_x, g_y)$:

$$\begin{aligned} \hat{n}_x &= g_x/\left\|\nabla\phi\right\|, \quad \hat{n}_y = g_y/\left\|\nabla\phi\right\|, \\ \psi_{\mathrm{prop}} &= \mathrm{atan2}(\hat{n}_x, \hat{n}_y) \mod 360^\circ, \\ \delta &= (\psi_{\mathrm{prop}} - \theta_{\max}) \mod 360^\circ, \\ \theta &= \min(\delta,\;360^\circ - \delta). \end{aligned} \tag{31}$$

4 Crown Fire (Van Wagner 1977)

Crown fire initiates when the surface fireline intensity exceeds a critical threshold:

$$I_{\text{crit}} = \big(0.01\, h_{\text{CBH}}\,(460 + 2590)\big)^{1.5} \quad [\text{kW/m}], \tag{32}$$

$$I_{\text{surf}} = I_R\, \xi\, R_{\text{dir}} \times 60,$$

where h_{CBH} is canopy base height (m).

Crown ROS (Rothermel 1991):

$$R_{\text{crown}} = \frac{11.02\, U_{10}^{0.854}\, \rho_c^{0.19}}{60} \quad [\text{m/s}], \tag{33}$$

where ρ_c is canopy bulk density (kg/m^3).

Effective ROS:

$$R_{\text{eff}} = \begin{cases} \max(R_{\text{dir}},\, R_{\text{crown}}) & I_{\text{surf}} > I_{\text{crit}}, \\ R_{\text{dir}} & \text{otherwise.} \end{cases} \tag{34}$$

5 Level-Set Fire Propagation

The fire front is implicitly represented by $\phi(\mathbf{x}, t)$, the signed distance function, where $\phi < 0$ inside the fire perimeter and $\phi > 0$ outside. The Hamilton-Jacobi PDE is:

$$\frac{\partial \phi}{\partial t} + R(\nabla\phi / \|\nabla\phi\|)\, \|\nabla\phi\| = 0. \tag{35}$$

5.1 Godunov upwind discretisation

For expanding fronts (ϕ decreasing), the upwind approximation is:

$$D_x^- = (\phi_{i,j} - \phi_{i-1,j})/\Delta x, \quad D_x^+ = (\phi_{i+1,j} - \phi_{i,j})/\Delta x,$$
$$D_y^- = (\phi_{i,j} - \phi_{i,j-1})/\Delta x, \quad D_y^+ = (\phi_{i,j+1} - \phi_{i,j})/\Delta x,$$
$$|\nabla\phi|_{\text{up}} = \Big[\max(D_x^-, 0)^2 + \max(-D_x^+, 0)^2 + \max(D_y^-, 0)^2 + \max(-D_y^+, 0)^2\Big]^{1/2} \tag{36}$$

For efficiency, replicate-padded central differences are used to compute the gradient components needed for the directional ROS:

$$g_x = \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta x}, \quad g_y = \frac{\phi_{i,j+1} - \phi_{i,j-1}}{2\Delta x}. \tag{37}$$

5.2 Time integration

A forward-Euler step with a CFL-adaptive timestep:

$$\phi^{n+1} = \phi^n - \Delta t\, R(\theta)\, |\nabla\phi|_{\text{up}}, \tag{38}$$

$$\Delta t = \min\Big(\nu\, \frac{\Delta x}{R_{\max}},\ 300\ \text{s}\Big), \tag{39}$$

where $\nu \in [0.4,\, 0.8]$ is the CFL safety factor and $R_{\max} = \max(R_{\text{head}}, R_{\text{crown}})$ is computed once per ensemble member before the loop.

5.3 Arrival time recording

Cell (i, j) is marked as ignited at the first timestep the front crosses zero. The exact crossing time is found by linear interpolation:

$$t_{\text{arr}}^{i,j} = t^n + \Delta t\, \frac{\phi_{i,j}^n}{\phi_{i,j}^n - \phi_{i,j}^{n+1} + \varepsilon}, \quad \phi^n > 0,\ \phi^{n+1} \le 0. \tag{40}$$

5.4 SDF redistancing

Every $K = 20$ timesteps the SDF is restored to a unit-gradient field via pseudo-time iterations (Sussman, Smereka & Osher 1994):

$$s_{i,j} = \phi_{i,j} / \sqrt{\phi_{i,j}^2 + \Delta x^2}, \tag{41}$$

$$\phi^{\tau+1} = \phi^\tau - \Delta\tau\, s\,(|\nabla\phi^\tau| - 1), \tag{42}$$

$$\Delta\tau = 0.5\, \Delta x, \tag{43}$$

applied for 5 iterations on cells with $|\phi| < 3\Delta x$.

6 Ensemble Fire State

The N-member ensemble provides a non-parametric distribution of arrival times. Members share GP-mean environmental conditions but have independent perturbations drawn from the GP posterior.

For member n, let $m^{(n)}$, $s^{(n)}$, $d^{(n)}$ be FMC, wind speed, and wind direction fields drawn from:

$$m^{(n)} \sim \mathcal{N}\big(\hat{m},\, \Sigma_m\big), \tag{44}$$

$$s^{(n)} \sim \mathcal{N}\big(\hat{s},\, \Sigma_s\big), \tag{45}$$

$$d^{(n)} \sim \mathcal{N}\big(\hat{d},\, \Sigma_d\big) \mod 360^\circ, \tag{46}$$

where Σ are diagonal matrices of GP posterior variances.

The ensemble arrival time field is:

$$T^{(n)}(r,c) = \begin{cases} t_{\text{arr}}^{(n)}(r,c) & \text{cell ignited within horizon,} \\ 2T_{\text{hor}} & \text{otherwise (sentinel),} \end{cases} \tag{47}$$

where T_{hor} is the planning horizon in minutes.

Summary statistics:

$$p_{\text{burn}}(r,c) = \frac{1}{N}\sum_{n=1}^{N} \mathbb{1}\big[T^{(n)} < 0.9 \times 2T_{\text{hor}}\big], \tag{48}$$

$$\bar{T}(r,c) = \frac{1}{|\mathcal{B}|}\sum_{n\in\mathcal{B}} T^{(n)}, \tag{49}$$

$$\sigma_T^2(r,c) = \frac{1}{N}\sum_{n=1}^{N}\big(T^{(n)} - \bar{T}\big)^2, \tag{50}$$

where $\mathcal{B} = \{n : T^{(n)} < 0.9 \times 2T_{\text{hor}}\}$.

7 Fire State Estimation

7.1 Arrival time reconstruction via fast marching

Given a set of fire-detection observations $\mathcal{O} = \{(r_k, c_k, t_k, q_k)\}$ (location, time, is-fire flag), the arrival field is initialised at observed fire cells: $T_{\text{arr}}(r_k, c_k) = t_k$.

An 8-connected Dijkstra fast march then fills unobserved cells. For a neighbour (r', c') at Euclidean distance factor ℓ:

$$T_{\text{arr}}(r',c') \leftarrow \min\Big(T_{\text{arr}}(r',c'),\ T_{\text{arr}}(r,c) + \frac{\ell\,\Delta x}{R_0(r',c')}\Big), \tag{51}$$

where R_0 is the scalar Rothermel ROS at GP-mean conditions ($\ell = 1$ for cardinal, $\ell = \sqrt{2}$ for diagonal neighbours).

7.2 Arrival uncertainty

$$d_k(r,c) = \Delta x \cdot \text{EDT}(\sim \text{obs mask}), \tag{52}$$

$$\sigma_{\text{sp}}(r,c) = 30 + 0.6\, d_k \quad [\text{s}], \tag{53}$$

$$\sigma_{\text{tp}}(r,c) = (t_{\text{now}} - t_{\text{obs}})\,\frac{\hat{R}_{95}}{\hat{R}_{95} + \varepsilon}, \tag{54}$$

$$\sigma_T(r,c) = \sqrt{\sigma_{\text{sp}}^2 + \sigma_{\text{tp}}^2}, \tag{55}$$

where $\hat{R}_{95}$ is the 95th-percentile ROS over burnable cells.

7.3 Ensemble initialisation

Each member arrival time is perturbed by a spatially correlated field $\xi^{(n)}$ (zero mean, unit variance, correlation length $L_c = 500$ m):

$$T_{\text{init}}^{(n)}(r,c) = T_{\text{arr}}(r,c) + \xi^{(n)}(r,c)\,\sigma_T(r,c), \tag{56}$$

with deep-interior ($T < T_{\text{min}} + 3600$) and far-exterior ($T > 0.9T_{\text{max}}$) cells clamped to their deterministic values.

The correlated field is generated via spectral filtering:

$$\begin{aligned} \tilde{Z} &= \mathcal{F}\big[\varepsilon\big], \quad \varepsilon \sim \mathcal{N}(0, I), \\ S(\mathbf{k}) &= \exp\big(-\tfrac{1}{2}\,\|\mathbf{k}\|^2\, L_c^2\big), \\ \xi &= \mathcal{F}^{-1}\big[\tilde{Z}\sqrt{S}\big]/\text{std}(\cdot). \end{aligned} \tag{57}$$

7.4 Particle filter update

Between initialisation and hard-reset events, members are reweighted using new fire observations. For member n and observation (r_k, c_k, q_k):

$$w_n \leftarrow w_n \times \begin{cases} c_k & q_k = \text{fire},\ T^{(n)}(r_k,c_k) < t_{\text{now}}, \\ 1 - c_k & q_k = \text{no-fire},\ T^{(n)} < t_{\text{now}}, \\ 1 - c_k & q_k = \text{fire},\ T^{(n)} \geq t_{\text{now}}, \\ c_k & q_k = \text{no-fire},\ T^{(n)} \geq t_{\text{now}}, \end{cases} \tag{58}$$

where $c_k \in [0,1]$ is detection confidence. Weights are normalised, and systematic resampling is applied when the effective sample size $N_{\text{eff}} = 1/\sum_n w_n^2 < N/2$.

8 Environmental GP Prior

8.1 Observation types

Each observation y_i at location (r_i, c_i, t_i) belongs to one of FMC, wind speed, or wind direction. Drone-camera FMC is modelled as spectrally derived, while RAWS stations provide all three variables.

8.2 Terrain-aware Matérn kernel

Feature vectors include spatial and terrain information:

$$\mathbf{x} = \big[\Delta\mathrm{N},\ \Delta\mathrm{E},\ z,\ \psi_{\mathrm{asp}}\big]^\top, \tag{59}$$

where $(\Delta\mathrm{N}, \Delta\mathrm{E})$ are northing/easting offsets, z is elevation, and ψ_{asp} is aspect.

Terrain-aware distance:

$$d(\mathbf{x}, \mathbf{x}') = \big\|\mathbf{x}_{1:2} - \mathbf{x}'_{1:2}\big\| + \alpha|z - z'| + \beta \min\big(|\psi - \psi'|,\ 360^\circ - |\psi - \psi'|\big). \tag{60}$$

Matérn $\nu = 3/2$ kernel:

$$u = \sqrt{3}\, d/\ell, \tag{61}$$

$$k(\mathbf{x}, \mathbf{x}') = C\,(1 + u)\, e^{-u}, \tag{62}$$

with amplitude C and length scale ℓ optimised per-variable.

8.3 GP posterior

Given n training points $\{(\mathbf{x}_i, y_i)\}$ with residuals $\tilde{y}_i = y_i - \mu_{\mathrm{prior}}(\mathbf{x}_i)$:

$$\mathbf{K}_{**} + \mathbf{R} = \mathbf{L}\mathbf{L}^\top, \quad R_{ii} = \sigma^2_{\mathrm{obs},i}, \tag{63}$$

$$\hat{\mu}(\mathbf{x}) = \mu_{\mathrm{prior}} + \mathbf{k}_*^\top (\mathbf{L}^\top)^{-1}\mathbf{L}^{-1}\tilde{\mathbf{y}}, \tag{64}$$

$$\hat{\sigma}^2(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - \mathbf{k}_*^\top (\mathbf{L}^\top)^{-1}\mathbf{L}^{-1}\mathbf{k}_*, \tag{65}$$

where $\mathbf{k}_* = [k(\mathbf{x}, \mathbf{x}_i)]_i$.

Wind direction is modelled with a circular GP: residuals are computed on the sin / cos representation and the circular mean is reconstructed via atan2.

8.4 Nelson FMC prior mean

The GP prior mean μ_{prior} for FMC is the Nelson (2000) equilibrium moisture content, which depends on relative humidity H (fraction), ambient temperature T (°C), elevation z, and hour of day.

Fosberg-Deeming three-segment EMC:

$$\mathrm{EMC}(\%) = \begin{cases} 0.03229 + 0.281073H' - 0.000578H'T & H' < 10, \\ 2.22749 + 0.160107H' - 0.01478T & 10 \le H' < 50, \\ 21.0606 + 0.005565H'^2 - 0.00035H'T - 0.00199T & H' \ge 50, \end{cases} \tag{66}$$

where $H' = 100H$.

Lapse-rate correction ($\Gamma_L = -0.0065$ °C/m):

$$T_{\mathrm{loc}} = T - \Gamma_L(z - z_{\mathrm{ref}}). \tag{67}$$

Solar drying uses the cosine of incidence angle:

$$\cos\iota = \sin\alpha_\odot \cos\alpha_s + \cos\alpha_\odot \sin\alpha_s \cos(\psi_\odot - \psi_s), \tag{68}$$

where $\alpha_\odot$ is solar elevation, α_s is terrain slope, and $\psi_\odot, \psi_s$ are solar and slope azimuths.

Canopy attenuation follows Beer–Lambert:

$$\tau = e^{-\kappa\, c_c}, \tag{69}$$

where c_c is fractional canopy cover and κ is an extinction coefficient. The final FMC prior is:

$$m_{\mathrm{prior}} = \mathrm{EMC} \times (1 - w_\odot \cos\iota\, \tau) \times \mathrm{elev_factor}, \tag{70}$$

clipped to $[m_{\min}, m_{\max}]$.

9 Ensemble Kalman Filter

Drone observations are assimilated into the GP via an ensemble Kalman filter (EnKF) operating on the GP state (FMC, wind speed, wind direction).

9.1 Observation aggregation

Multiple observations at nearby locations are precision-weighted:

$$\hat{m}_{\mathrm{agg}} = \frac{\sum_i \sigma_{m,i}^{-2}\, m_i}{\sum_i \sigma_{m,i}^{-2}}, \quad \hat{\sigma}_{\mathrm{agg}} = \Big(\sum_i \sigma_{m,i}^{-2}\Big)^{-1/2}. \tag{71}$$

Wind direction uses circular precision weighting:

$$\bar{s} = \frac{\sum_i w_i \sin d_i}{\sum_i w_i}, \quad \bar{c} = \frac{\sum_i w_i \cos d_i}{\sum_i w_i}, \qquad \hat{d}_{\mathrm{agg}} = \mathrm{atan2}(\bar{s}, \bar{c}) \mod 360^\circ. \tag{72}$$

9.2 EnKF update

Let $\mathbf{X} \in \mathbb{R}^{N\times D}$ be the GP state ensemble, $\mathbf{y} \in \mathbb{R}^p$ the observation vector, and $\mathbf{H} \in \mathbb{R}^{p\times D}$ the linear observation operator (index gather at observed cells).

Covariance inflation:

$$\mathbf{A} = \lambda(\mathbf{X} - \bar{\mathbf{X}}), \quad \mathbf{X}_{\text{inf}} = \bar{\mathbf{X}} + \mathbf{A}, \tag{73}$$

where $\lambda > 1$ is the inflation factor.

Ensemble covariances:

$$\mathbf{PH}^\top = \frac{\mathbf{A}^\top(\mathbf{HA})}{N-1}, \tag{74}$$

$$\mathbf{HPH}^\top = \frac{(\mathbf{HA})^\top(\mathbf{HA})}{N-1}, \tag{75}$$

$$\mathbf{K} = \mathbf{PH}^\top(\mathbf{HPH}^\top + \mathbf{R})^{-1}, \tag{76}$$

where $\mathbf{R} = \text{diag}(\sigma^2_{\text{obs},1}, \ldots)$.

Gaspari–Cohn localisation:

$$K_{ij} \leftarrow K_{ij}\, \rho\big(d(i,j)/r_{\text{loc}}\big), \tag{77}$$

where ρ is the Gaspari–Cohn taper function and r_{loc} is the localisation radius.

Perturbed-observations update:

$$\begin{aligned} \boldsymbol{\varepsilon}^{(n)} &\sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \\ \mathbf{x}^{(n)\prime} &= \mathbf{x}^{(n)}_{\text{inf}} + \mathbf{K}\big(\mathbf{y} + \boldsymbol{\varepsilon}^{(n)} - \mathbf{H}\mathbf{x}^{(n)}_{\text{inf}}\big). \end{aligned} \tag{78}$$

The updated ensemble is used to refit the GP for the next cycle.

10 Information Field

The information field $w(r,c)$ quantifies the expected utility of a drone observation at cell (r,c).

10.1 Binary entropy

$$H(p) = -(p\log_2 p + (1-p)\log_2(1-p)), \quad p \in (0,1), \tag{79}$$

with $H(0) = H(1) = 0$ and $H(0.5) = 1$.

10.2 Bimodality score

The burn probability $p_b(r,c) = N^{-1}\sum_n \mathbb{1}[T^{(n)} < T_0]$ (with $T_0 = 2T_{\text{hor}}$ the unburned sentinel threshold) measures ensemble spread about the ignition/non-ignition boundary:

$$b(r,c) = H\big(p_b(r,c)\big). \tag{80}$$

10.3 Sensitivity to environmental variables

Let $\mathbf{a}^{(n)} \in \mathbb{R}^D$ be the flattened arrival times for member n, and $\mathbf{p}^{(n)}$ a perturbation field (FMC, wind speed, or wind direction). The pointwise Pearson correlation is:

$$\rho_{\text{sens}}(r,c) = \frac{\text{Cov}(a_{r,c}, p_{r,c})}{\sigma_a(r,c)\,\sigma_p(r,c) + \varepsilon}. \tag{81}$$

10.4 Observability degradation

Sensor performance degrades near an active fire front. Define the perimeter mask $\mathcal{P} = \{(r,c) : p_{\text{lo}} < p_b < p_{\text{hi}}\}$. Then:

$$\begin{aligned} d_{\mathcal{P}}(r,c) &= \Delta x \cdot \text{EDT}(\sim \mathcal{P}), \\ D(r,c) &= \min\big(d_{\mathcal{P}}(r,c)/\rho_{\text{deg}},\ 1\big), \end{aligned} \tag{82}$$

where ρ_{deg} is the degradation radius.

10.5 Combined information value

$$\begin{aligned} w(r,c) = \hat{\sigma}^2_m\, |\rho_{\text{sens},m}|\, D_m \\ + \hat{\sigma}^2_s\, |\rho_{\text{sens},s}|\, D_s \\ + \alpha_b\, b(r,c) \\ + \alpha_c\, H\big(p_{\text{crown}}(r,c)\big), \end{aligned} \tag{83}$$

where $\alpha_b, \alpha_c \geq 0$ are weighting coefficients and $p_{\text{crown}} = N^{-1}\sum_n \mathbb{1}[\text{type}^{(n)} = \text{crown}]$. Non-burnable cells (water, urban, rock; SB40 codes {91–99}) are masked to zero before path planning.

11 Information-Theoretic Drone Path Planning

11.1 Domain decomposition

The grid is partitioned into correlated domains via Felzenszwalb graph-based segmentation applied to normalised terrain features (elevation 1.0, slope 0.5, aspect 1.5×, fuel 2.0, canopy 0.8):

$$\mathcal{D} = \text{Felzenszwalb}\big(\mathbf{F}, \lambda_s, \sigma_{\text{blur}}, n_{\min}\big), \tag{84}$$

where $\lambda_s = L_c/(2\Delta x)$, $\sigma_{\text{blur}} = 0.8$, and $n_{\min}$ is the minimum domain size in cells.

Each domain $d \in \mathcal{D}$ has a representative cell $c^*(d) = \arg\max_{(r,c)\in d} w(r,c)$.

11.2 Domain graph

An edge between adjacent domains (d_i, d_j) is weighted by:

$$\ell_{ij} = \|c^*(d_i) - c^*(d_j)\| \; \Delta x \quad [\text{m}], \tag{85}$$

$$\rho_{ij} = \exp\big(-\|\mathbf{f}_i - \mathbf{f}_j\|\big) \quad (\text{cross-correlation proxy}), \tag{86}$$

where $\mathbf{f}_i$ is the normalised feature vector at $c^*(d_i)$.

Single-source Dijkstra from the staging area gives the minimum-distance path $\ell^*(d)$ to each domain.

11.3 Greedy orienteering

The drone starts at the staging area with range budget $B = v_{\text{drone}} T_{\text{end}}$ m. At each step, the next domain is:

$$d^* = \arg\max_{d \text{ unvisited}} \frac{w(d) + g_{ij}}{\ell_{ij} + 1}, \tag{87}$$

subject to the *return feasibility* constraint:

$$\ell_{\text{traveled}} + \ell_{ij} + \ell^*(d) + \ell_{\text{safe}} \le B, \tag{88}$$

where ℓ_{safe} is a 10% safety buffer.

11.4 GP-conditioned greedy update

Within the first-cycle budget, each selected domain hypothetically reduces GP variance. For domain d_i just selected, the posterior variance at any other domain d_j is updated via:

$$\hat{\sigma}^{2\prime}(d_j) = \hat{\sigma}^2(d_j) - \frac{k(d_j, d_i)^2}{k(d_i, d_i) + \sigma^2_{\text{sensor}}}, \tag{89}$$

and $w(d_j)$ is rescaled by $\hat{\sigma}^{2\prime}(d_j)/\hat{\sigma}^2(d_j)$.

11.5 Multi-drone deconfliction

For K drones, domains already claimed by an earlier drone are removed from the feasible set. Each drone plans sequentially on the residual information field, so the joint plan maximises:

$$\max_{\pi_1,\ldots,\pi_K} \sum_{k=1}^{K} \sum_{d\in\pi_k} w'(d), \tag{90}$$

where w' is updated after each drone plans.

12 Evaluation Metrics

12.1 Continuous Ranked Probability Score

Given ensemble members $\{x_1, \ldots, x_N\}$ and scalar truth y (both in minutes), the CRPS is:

$$\text{CRPS} = \mathbb{E}|X - y| - \tfrac{1}{2}\mathbb{E}|X - X'|, \tag{91}$$

computed via the sorted-member identity:

$$\mathbb{E}|X - y| \approx \frac{1}{N}\sum_{n=1}^{N} |x_n - y|, \tag{92}$$

$$\tfrac{1}{2}\mathbb{E}|X - X'| \approx \frac{1}{N^2}\sum_{n=1}^{N} x_{(n)}(2n - N - 1), \tag{93}$$

where $x_{(n)}$ are order statistics. CRPS is averaged over cells burned within the planning horizon.

12.2 Spread-skill ratio

$$\text{SSR} = \frac{\sqrt{\overline{\sigma^2_T}}}{\max(\text{RMSE}_{\bar{T}},\, \varepsilon)}, \tag{94}$$

where $\overline{\sigma^2_T}$ is the mean ensemble variance and $\text{RMSE}_{\bar{T}}$ is the RMSE of the ensemble mean against ground truth. SSR ≈ 1 indicates a calibrated ensemble.

12.3 Placement Error Rate (PERR)

$$\text{PERR} = 1 - \frac{|\mathcal{S}_{\text{pred}} \cap \mathcal{S}_{\text{opt}}|}{|\mathcal{S}_{\text{pred}} \cup \mathcal{S}_{\text{opt}}|}, \tag{95}$$

the Jaccard distance between predicted and oracle-optimal waypoint sets.

Symbol	Description	Units
Δx	Grid resolution	m
ϕ	Signed-distance field	m
R_{head}	Head-fire ROS	m/s
I_R	Reaction intensity	kW/m^2
ϕ_w, ϕ_s	Wind/slope amplification	–
e	Ellipse eccentricity	–
$T^{(n)}$	Member arrival time	min
p_b	Burn probability	–
w	Information value	–
$\mathbf{K}$	Kalman gain	–
ℓ	GP length scale	m
CRPS	Continuous ranked probability score	min
SSR	Spread-skill ratio	–

13 Notation Summary

References

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